Bluetooth Low Energy

1 Acronyms and terms

- BR/EDR = Basic Rate / Enhanced Data Rate = Bluetooth Classic.
- PDU: Protocol Data Unit, packet or message

2 Links

- Specifications and documents

3 The Bluetooth Low Energy Primer

Document Version: 1.2.0, 15th March 2024, M.Woolley and I.Anees, Bluetooth SIG.

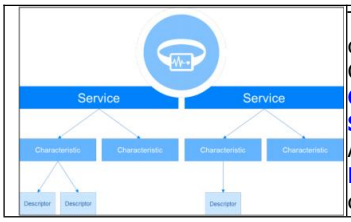
3.1 The ATT Protocol

Roles in ATT, a device is client, the other is server. The server exposes compisite data items called attributes, organized in an **attribute table**. Each attribute has:

- Handle: UUID
- Type expressed as UUID
- Permission R, W, RW.
- Value

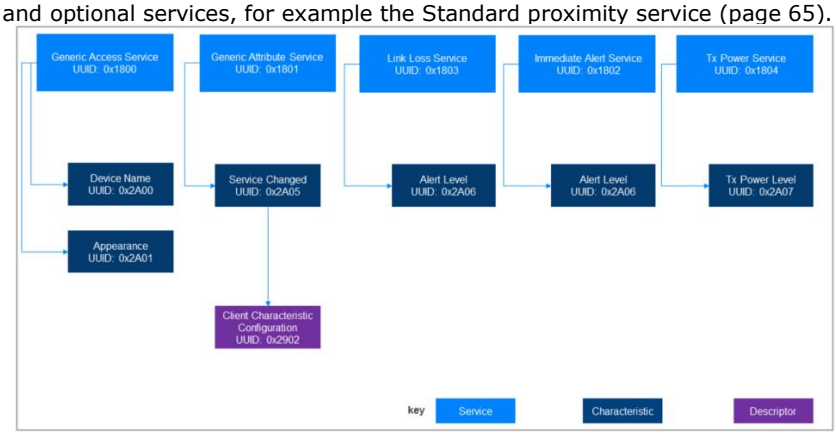
The GATT defines how attributes compose servics, characteristics and descriptors.
There is Basic ATT and enhanced ATT. ATT defines commands e.g. write, initiated by the client. The server can send insolicited *notifications* and *indications*. Notifications do not require acknowledgement from client. Indications require acknowledgement from the client. The ATT follows a sequential transaction model: no other PDU can sent until the current one has received response or timeout expired. In transactions the timeout is **30s**.

3.2 The Generic Attribute Profile (GATT)



The GATT defines data types on top of the ATT. GATT also defines procedures. Application level API(s) are mapped to GATT procedures.
Characteristics: individual data items = ATT attributes.
Services group characteristics and provide context.
A characteristic may belong to more than one service.
Descriptors contain metadata that belong to characteristics.

A characteristic can have zero or more attached descriptors.
Example: a client can request notification, priodically or if a value changes. Notifications are only sent to the peer that requested them. So notifications are managed by setting a descriptor related to the notified characteristic.
The GATT devies the roles: **server** and **client**.
Two special services are mandatory in all GATT servers: the generic access service and the generic attribute service.
GATT procedures cover: service discovery, characteristic discovery, descriptor discovery, reading and writing characteristic values, notifications and indications.
Profiles define mandatory and optional services, for example the Standard proximity service (page 65).



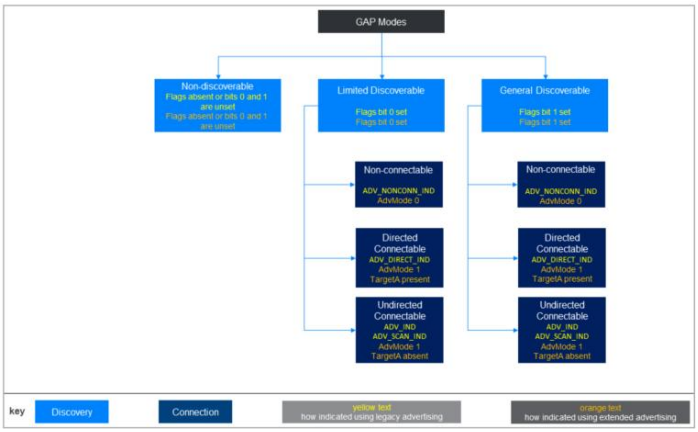
A service can mix Bluetooth SIG defined (I.e. standard) and custom attributes.

3.3 The Generic Access Profile (GAP)

(Page 67) The base of GAP are advertising packets and their receipt through scanning. There are different advertising and scanning packets. Roles:

- Broadcster: transmits advertising data that is connectionless. It does not accept connections.
- Observer: receives advertising packets. Receives broadcast data bit it does not attempt to connect to the Broadcaster.
- Peripheral: it can be connected to a Central device.
- Central: it can initiate the establishment of a connection with a peripheral.

Advertising can be undirected, or directed to a specific device.



(Page 69)

Certain advertising PDU are scannable. It means that the receivinf device can respond with a scan request PDU to request more advertising data.

3.4 The Security Manager Protocol (SMP)

The SMP is part of the security manager component. The SMP supports the execution of security related procedures: pairing, bonding and key distribution.

3.5 Security in Bluetooth LE

4 Develop your own Bluetooth low energy applications

Book *by Koen Vervloesem -- 2023 -- PUBLITR ELEKTOR*, 258 pages.
Read up to page: 100
In C++, the book describes the NimBLE Arduino library. Install the **nRF Connect mobile app**.
BLE devices only allow a connection to one device [34].

4.1 Chp.3 Broadcasting data with advertisements

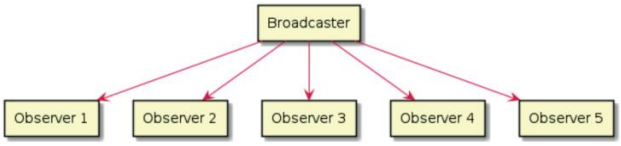
Communication without a connection: called advertising. Mostly boadcasting: any BLE device can receive broadcast packets.
Advertisement packets are fundamental for device discovery. The broadcst sender does not know if any receiver is listening. Advertisement is an unreliable operation.
For scanning, use a powerful platform, for advertising a less powerful device.

4.1.1 Device roles: the GAP roles

The Generic Access Profile (GAP) defines four roles:
Broadcaster and **observer**.

- Broadcaster: sends non-connectable advertising packets to any device in the neighborhood
- Observer: scans non-connectable advertising packets send by broadcasters.

One-to-many relation between broadcaster and observer(s)



Active scan. The scanner can send a SCAN-REQ advertisement directed to a broadcaster to request extra information, directed to the broadcaster. The broadcaster replies with SCAN_RSP. Called scan response data.
Passive Scan: the scanner only listens to advertising packets, it does not send SCAN_REQ-packets.
Central and **peripheral** => master and slave

- Peripheral: sends connectable advertising packets. Once connected, the peripheral becomes slave to the central
- Central: initiates a connection to a peripheral. Once connected, it is the master in the link layer.

A BLE device can have multiple GAP roles at the same time.

BLE uses 40 2MHz channels: three channels are advertising channels: 37, 38, and 39. The other channels are used by connected channels

As advertising packet can carry up to 31 bytes of data. Page 46: advertising packet structure
BLE transmit data in little endian format.

BLE central must find devices in the neighborhood.

4.1.2 Bluetooth addresses

Every BLE device has a 48-bit Bluetooth address.

- **Public addresses**: globally unique addresses registered with the IEEE. 48bit extended unique identifier: 24 bit company ID and 24 device ID assigned by the organization.
- **Random addresses**: programmed into the device or generated by the application. It is not guaranteed to be unique. It

does not require registration with the IEEE. A random address can be:

- **Static address**: randomly chosen by the manufacturer or developer
- **Private address**: it changes periodically, e.g. every 15 minutes. Moreover, private addresses can be:
  - Non-resolvable addresses: it does not allow anyone to track the device
  - Resolvable address: only trusted devices can track the device. The device identity can be resolved using a shared key.

In Linux it is possible to generate random UUID with the **`uuidgen`** command.

4.2 Connections and services

Peripheral sends connectable advertising packets. Once connected, the peripheral has slave role in the link layer. The central initiates the connection. Once connected, it becomes the master. Connections are ont-to-one method of communication, in both directions. Both devices can send data to th other device.
Peripherals usually stop sending advertising packets when connected. However, since BLE 4.1, a peripheral can be connected with multiple centralsat the same time.
Master/slave, Central/peripheral are GAP roles.

GATT roles are client and server.

4.2.1 The GATT

A GATT server is a database or table of attributes
Each piece of data is described by an attribute, which has:

- Handle
- Type
- Permissions
- Value: maximum length is 512 bytes.

4.2.1.1 Services, characteristics, descriptors.

A GATT server has:

- One or more services
- Each service groups one or more related characteristics
- Each characteristic can have zero or more descriptors

5 Getting Started with Bluetooth Low Energy

5.1 Introduction

[KT14] BLE is a radio standard with the minimum power consumption. [3] BLE 4.0 was introduced in 2010, while BLE 4.1 was released in December 2013. The classic Bluetooth and BLE are not directly compatible: [4] Pre-4.0 Bluetooth devices cannot communicate with BLE devices, and BLE single-mode devices cannot communicate with BT Classic devices. 4.x dual mode devices can communicate with both BLE and BT-Classic devices. BR/EDR is the BT Classic.

[5] Basic building blocks of any BT device:

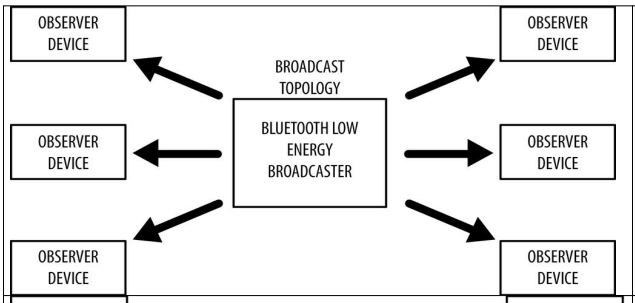
- Application
- Host: the upper layers of the BT protocol stack
- Controller: the lower layers of the stack.

The specification includes the Host-Controller Interface HCI, to allow interoperability between hosts and controllers by different vendors.

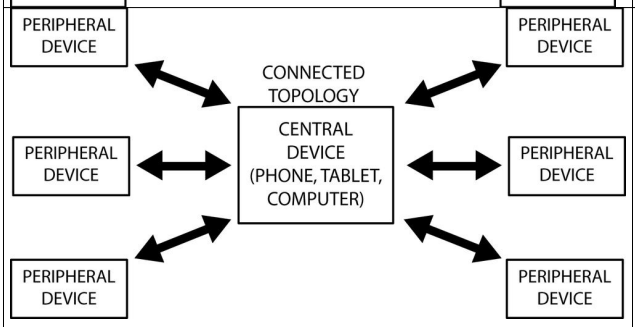
Limitations of the BLE 4.1: **theoretical data throughtput** is 1Mbps. When a connection is active, there is a connection interval between two consecutive connection events, in the interval between 7.5ms and 4s. Between connection events, the device is in power save. Actual throughput is about 5-10kB/s [8]

Operating range: the maximum theoretical is `30 meters`, but the typical operating range is `2 to 5 meters`.

Network topology



Broadcasting / Observing. Broadcasting is connectionless, data is sent one-way to any observer that can pick the received data [9]. The Broadcaster sends non-connectible *advertising* packets
The Observer scans the preset frequencies to receive advertising packets.
It is the only way to implement a one-transmitter / multiple-receivers.
The standard advertising packet payload is 31 bytes. An additional 31bytes Scan Response can be sent, so the maximum payload is 62 bytes [10]. The Scan Response is requested by the observer and send by the broadcaster.



Connection The Peripheral (slave) sends periodic connectable advertising packets and accepts incoming connections. Once connected, the slave follows the timing dictated by the master, the slave stops advertising.
The Central (master) scans the preset frequencies for advertising packets and initiates a connection. Once connected, the master manages the timing and initiates the periodic data exchanges.
While the connection is active, data can be sent by any device during each connection event.
[11] Starting from BLE 4.1, no limitations on connections and roles.
Connection allows to save battery power.

5.1.1 Protocols and profiles

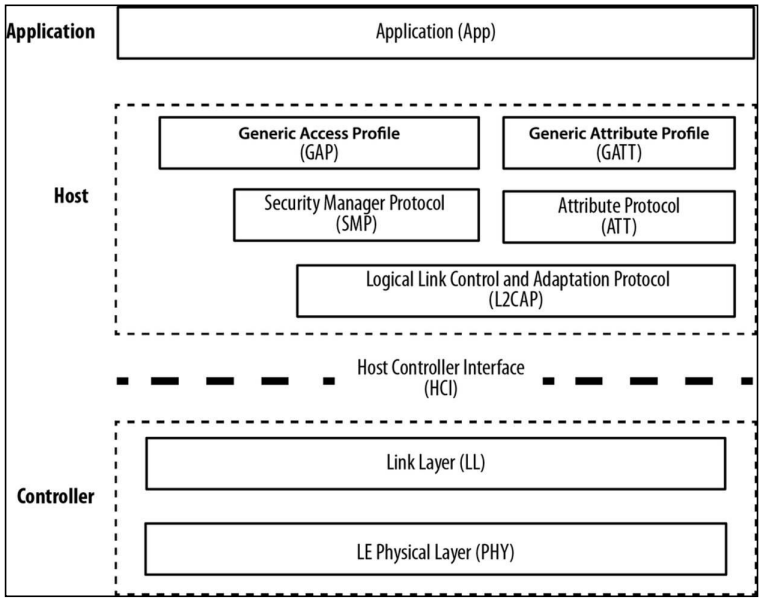
Protocols allow data to be exchanged between peers. Profiles define how protocols should be used to achieve a particular goal.

Generic profiles are defined by the specification.

- **Generic Access Profile (GAP)** the usage model for the lower-level radio protocols, it allows devices to broadcast data, discover devices, establish connections, etc. It is the top-most control layer of the BLE
- **Generic Attribute Profile (GATT)**: it defines a data model to exchange data between devices. It is the top-most data layer of the BLE

Use-case specific profiles: they are limited to GATT-based profiles. Find-Me, Proximity, Cycling Speed and Cadence, etc.

5.2 Protocol Basics



<table>
<tr><td>PHY</td><td>2.4GHz radio, frequency hopping spread spectrum</td></tr>
<tr><td>LL</td><td>
The LL defines the roles: Advertiser (sends adv packets), Scanner (scans adv packets), Master (initiates the connection), Slave (accepts the connection and follows the master's timing). Advertiser/scanner when no connection, master/slave when the connection is established.
Bluetooth Device Address (BDA): 48-bit, similar to MAC address. Two types: [19]

public device address: factory programmed, never changed
random device address: dynamically generated

Packets: one packet format and two packet types (advertising and data)

Advertising: 31 bytes payload, blindly broadcasted. Adv.packets are send cyclically to three frequencies, picked when the sender and the receiver overlap on the same frequency Adv.packets are sent every period, from 20ms to 10,24s.
Passive scanning: the scanner listens, the advertiser is not aware if the sent data was received
Active Scanning: the scanner issues a Scan-Request when it receives the adv.packet. The advertiser then sends a Scan-Response, with additional data. So, the advertiser can send additional data, but the scanner cannot.
Advertising packet types can be classified according to three properties:

Connectability: can the scanner initiate a connection upon reception of the adv.packet? If No, then broadcast only.
Scannability: can the scanner issue a Scan-Request when it receives the adv.packet?
Directability: a directed packet contains the advertiser and scanner BDA in its payload, no user data, so it is only connectable. Undirected packets are not directed to any particular scanner and they contain data in the payload.

<table>
<tr><th>Advertising Packet Type</th><th>Connectable</th><th>Scannable</th><th>Directed</th><th>GAP Name</th></tr>
<tr><td>ADV_IND</td><td>Yes</td><td>Yes</td><td>No</td><td>Connectable Undirected Advertising</td></tr>
<tr><td>ADV_DIRECT_IND</td><td>Yes</td><td>No</td><td>Yes</td><td>Connectable Directed Advertising</td></tr>
<tr><td>ADV_NONCONN_IND</td><td>No</td><td>No</td><td>No</td><td>Non-connectable Undirected Advertising</td></tr>
<tr><td>ADV_SCAN_IND</td><td>No</td><td>Yes</td><td>No</td><td>Scannable Undirected Advertising</td></tr>
</table>
[[KT14], page 22]

Connection: the master scans advertisers that accept connection requests (connectability), it found, the master sends a connection request, containing three connection parameters:

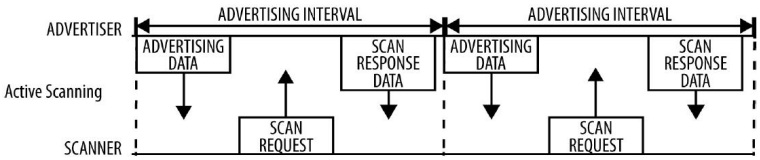
Connection interval: the period between two consecutive connection events [7.5ms, 4s]
Slave latency: the number of connection events that the slave can skip without risking a disconnection
Connection supervision timeout: the maximum time between two valid data packets are received.

A connection is a sequence of data exchanges between slave and master. Each exchange is a connection event. The radio is idle between connection events. A Connection event include one or more transmissions of packets.
The LL also features a White List: scanner (resp. Advertiser) can drop packets from devices whose BDA is not in the White List. The white list can be used by the scanner of the advertiser.
Data packets: usable data payload is 27 bytes. LL is reliable, packets are CRC checked and possibly retransmitted.
LL procedures:

advertising
scanning
establishing connections
transmitting data
receiving data
changing connection parameters
encryption: the LL manages encryption transparently to the upper layers

</td></tr>
<tr><td>HCI</td><td>Host and Controller interact via a serial protocol</td></tr>
</table>

L2CAP	• A protocol multiplexer that takes multiple protocols from upper layers and encapsulates them in the BLE packet • fragmentation and recombination: upper layers can send large packets, but they are fragmented into the payload size (e.g.27 bytes) In BLE, L2CAP routes two main protocols Attribute Protocol (ATT) and Security Manager (SMP). Since version 4.1, L2CAP can create **user-defined channels** for high-throughput data transfer, without the complexity of the ATT. [25] Due to the L2CAP, the effective user data payload size is **`23 bytes`**.
ATT	[26] A simple client/server stateless protocol based on attributes presented by a device. Client/server roles regardless of being slave, master, advertiser, etc. The client requires data, the server replies. Server contains data organized as attributes. Each peer can act both as client and server. Client and server depend on the data flow direction [37], they are NOT roles. An attribute has a UUID, permissions and a value. ATT operations [26]: error handling, find information, read, write. An attribute has [26]: 16bit handle, UUID, permissions, value. The handle is used to access the attribute value.
SMP	
GATT	Builds over the ATT, adding hierarchy and data abstraction model on top of it. It is the backbone of BLE transfer. Data is encapsulated into services. Services consist in one or more characteristics.
GAP	Generic Access Profile. How devices interact with each other at a lower level.



Active scanning

5.3 The Generic Access Profile protocol (GAP)

5.3.1 Roles

The protocol defines different aspects of device interactions [35]. Note that GAP roles are unrelated to GATT roles (client / server).

Roles	Each device operates in one or more roles at the same time. Roles impose restrictions and behaviors. Broadcaster periodically sends adv.packets with data. Example a thermometer that broadcasts temperature readings Uses the LL advertiser role (LL=Link Layer). Observer listens for payload data in adv.packets from broadcasting peers. LL scanner role. Central is the initiator of connections. Allows devices into the network. It is the link layer Master. LL master. Peripheral role is the LL Slave. It sends adv.packets so that centrals find it and establish a connection.
Modes	A mode refines the concept of role. A mode is a state in which a device can switch for a certain amount of time to achieve a particular goal.
Procedures	A procedure is a sequence of actions to achieve a goal. A procedure is usually associated to a mode.
Security	GAP builds on top of the security manager.
Additional GAP formats	

[37] Example of the fitness tracker paired with the smartphone. The tracker is peripheral, the smartphone is central, but in the GATT the tracker, can be server (provides the heart-rate) or client (requests the exact time to the smartphone).

5.3.2 Procedures

Procedure	Applicable Role(s)	Applicable Peer Mode(s)
Observation	Observer	Broadcast
Limited discovery	Central	Limited discoverable
General discovery	Central	Limited and General discoverable
Name discovery	Peripheral, central	N/A
Any connection establishment	Central	Any connectable
Connection parameter update	Peripheral, central	N/A
Terminate connection	Peripheral, central	N/A

5.3.2.1 Broadcast mode and Observation procedure

[38] The Broadcaster does not know whether it packet was received by any peer. Once the Obserser has received a packet it does not know whether more packets will follow. Adv.packets contain `31 bytes` and actual valid user data. The most critical data must be placed in the adv.packet, not in the Scan-Response packet. A broadcaster-only device simply broadcasts data, while a peripheral stops advertising itself after a connection is established.

5.3.2.2 Discovery

The device discoverability is how the peripheral advertises itself to other devices. Discoverability is controlled by the optional field *Flags AD* in the adv.packet [39]. Discoverability modes of a peripheral are:

- Non-discoverable: a device does not want to be found by central peers. It can still send adv.packets, but it has to clear the General and Limited discoverable bits.
- Limited discoverable mode
- General discoverable mode

The Discovery procedures are Limited Discovery procedure and General discovery procedure.

5.3.2.3 Connection Establishment

A Central initiates a connection to a peripheral. Connection establishment modes are:

- Non connectable mode
- Directed connectable mode: the periperal can send `ADV_DIRECT_IND` advertising packets directed to a central peer, to execute a fast reconnect .
- Undirected connectable mode: the standard mode used by a peripheral to make itself connectable. The device sends `ADV_IND` adv.packets.

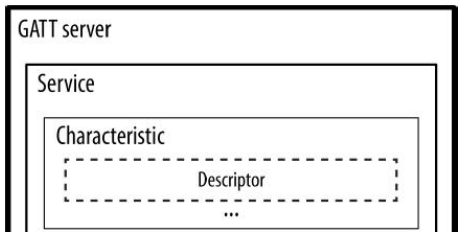
Connection establishment procedures[41]. A Central can connect regardless of the received adv.packet type (`ADV_IND` or `ADV_DIRECT_IND`)

- Auto-connection establishment procedure: the central populates a white-list and connect to the first available peripheral
- General connection establ. procedure: the central scans without a white-list, accepting all adv-packets. When a peripheral is chosen, the central connects to it, using the direct connection establlishment procedure.
- Selective conn.establ.procedure: same as general, but the central filters incoming adv.packets using a white-list.
- Direct connection establishment procedure: the host sends a direct message to a single device, identified by its BDA, without knowing if the target peripheral is available (the scanning step is skipped).

5.3.2.4 Additional GAP procedures [42]

- Name discovery procedure
- Connection parameters update
- Terminate connection

5.4 The GATT services and characteristics



Each standard BLE device has attributes organized into: Services, Characteristics, Descriptors.
One or more services, zero or more characteristics, zero or more descriptors. No attribute can live outside of this hierarchy [57].
Difference between definition and declaration. The declaration is a single attribute. Declarations have read-only permissions and no security.

[52] The GATT establishes how to exchange data over a BLE connection. The GATT is based on the ATT. GATT is the reference framework for all GATT-based profiles (use-cases). Data is organized in sections called **services**. Services group related pieces of user-data called **characteristics**. Roles:

- Client: the GATT client does not know anything in advance, so it has to inquire attributes performing service discovery
- Server: it is the ATT server.

UUID(s). An UUID is a 128-bit (16-bytes) number that is globally unique. BLE has 16-bits and 32-bits UUID(s). it is possible to reconstruct the full UUID from the shortened UUID [52]. It is possible to generate new UUID(s) using the link [UGEN]

5.4.1 Attributes

Attributes are the smalles piece of information that can contain user data (and metadata), defined by the ATT. GATT and ATT only work with attributes. Each attribute has information about itself and the actual data. The set of attributes can be displayed as a table [56].

Handle	A 16-bit identifier, unique on a particular GATT server. (`0x0000` is the invalid handle). A server provised a set of handles, they do not have to be contiguous. The client must use the discovery to determine valid handles.
Type	The type is a UUID. The UUID length can be 2, 4, or 16 bytes. The type determine the kind of data in the attribute value. It can be a standard UUID, or a vendor-specific UUID.
Permissions	• Access permissions: None / Readable / Writable / Readable and Writable • Encryption • Authorization
Value	No restrictions: it is a non-typed buffer. The maximum length is `512 bytes`.

5.4.2 Attributes and data hierarchy

GATT builds on top of the ATT and establishes a hierarchy to organize attributes.

GATT server -> Services -> Characteristics -> Descriptors.

- The attributes of a GATT server are grouped into **Services**.
- A **service** contains zero or more **characteristics**.
- A characteristic contains zero or more descriptors

All attributes in a GATT service belongs to one of the categories: service, characteristic, descriptor.

On the GATT hierarchy, differentiate between:

- **definition**:
- **declaration**: a single attribute, placed first within the definition, it intriduces the metadata about the data that follows. Declarations are read-only and without security.

5.4.2.1 Services

All attributes of a given service are grouped. Service definition: all attributes of the service, starting with the service declaration. [58].

Handle	Type	Permissions	Value	Value length
0xNNNN	$UUID_{primary\ service}$ or $UUID_{secondary\ service}$	Read Only	Service UUID	2, 4, or 16 bytes

There are primary and secondary services [58].

In the service declaration, the value is the service UUID.

A service can include other services. It is like a pointer or a reference to another object instance [58].

5.4.2.2 Characteristics

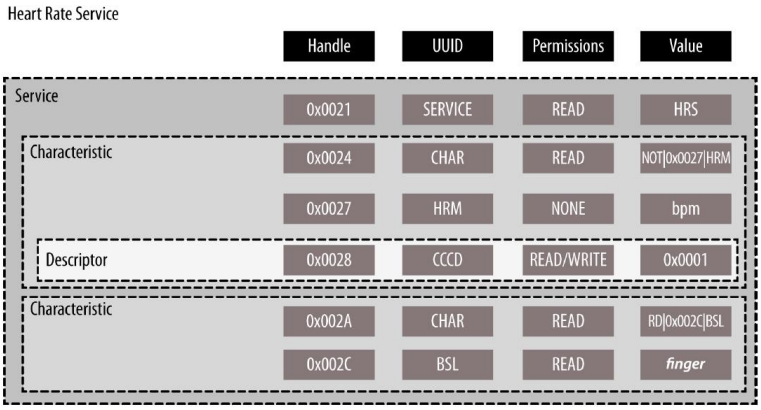
Characteristics are containers of user-data. They include the characteristic declaration and the characteristic value (the user data in its value field).
The characteristic definition contains:

- Characteristic declaration: the metadata
- Characteristic value: the actual data, in the value field.
- Additional descriptors can follow the charactestic value: they expand the metadata of the charact.declaration.

Charact.Declaration	Uses a standardized UUID (0x2803) to denote the beginning of a characteristic. Read-only permissions. It contains: • **Properties**: the operations and procedures that are allowed on this characteristic. Of interest are the Notify and Initiate properties. Length: 1 byte. • **Characteristic value handle**: the handle of the attribute containing the actual value of this characteristic. Length: 2 bytes. • **Characteristic UUID**. Either a standard UUID or a vendor specific UUID. 2, 4, or 16 bytes.
Charact.Value	[61] Characteristic value attributes no longer have the type of services of characteristic, but concrete types. The characteristic value has the same UUID as in the corresponding characteristic declaration.
Charact.Descriptors	A descriptor is a single attribute, with a standard UUID. The descriptors are optional, they expand the metadata of the characteristic declaration. There are two types of descriptors: • GATT-defined descriptors • Profile or vendor-defined descriptors

GATT-defined descriptors:

- Characteristic user description descriptor: a human readable description of the characteristic.
- Client Characteristic Configuration Descriptor (CCCD): enables/disables the server-initiated updates of the characteristic. Server-initiated update = the server asynchronously sends the updated characteristics. The CCCD is a two-bit fields enabling notifications and indications [63]. To enable notifications/indications for a characteristic, the client issues Write Request ATT to set the corresponding bit to 1 or 0. Moreover:
  - CCCD(s) are unique per connection.
  - Their value is preserved across connections to bounded devices.
- Characteristic presentation format descriptor.



Example: the HRM service

5.4.3 Advanced attribute concepts

Attribute caching. The specification recommends that a client caches the handles it is interested in. Values, are volatile and must not be cached. The server can communicate whether services changed [67].

GATT Attribute data in adv-packets. It is possible to include attributes hosted by a server inside adv-packets [67].

5.5 Links

https://www.jaredwolff.com/get-started-with-bluetooth-low-energy/
https://github.com/labapart/gattlib
https://git.kernel.org/pub/scm/bluetooth/bluez.git

6 Bluetooth Tutorial

[GC19] Wireless usage scenarios: Wireless Personal Area Net (**WPAN**), Wireless Lan (WLAN), Wireless WLAN (WWLAN).

WPAN scenarios are ad-hoc and spontaneous, wireless technology reduces complexity (removes the need for cables).

WLAN: a permanent wireless connection within a defined region.

What is Bluetooth[157]: a technology standard that uses short range radio links to replace cables. Key features: robustness, low complexity, low power, low cost.

Bluetooth characteristics [189]: BT operates in the unlicensed band `[2.402, 2.480] Ghz`, 79 channels. The range is 10 meters.

The BT radio system is optimized for mobility [189].

- Asynchronous Connectionless Link (ACL) for data transmission
- Synchronous Connection Oriented (SCO) for voice/audio transmission.

The gross BT rate is 1Mbps, while the effective rate is:

- Asymmetric: `721 kbps` and `57.6 kbps` in return
- Symmetric: `436.6 kbps` either side

6.1 The Bluetooth Architecture

[461]The Bluetooth Protocol Architecture is developed by the Bluetooth Special Interest Group (SIG), set-up in May 1998 [473].

[214] There is a master and one or more slave radios. Each BT radio has a fixed 48bit address **BD_ADDR**. Two to more BT devices form an ad-hoc network called **piconet**. All units in the same piconet share the same channel. A piconet contains a master and up to 7 slaves. Each device in the piconet is identified by a 3-bit device address.

- Only the master can initiate a communication link
- once the link is initiated, a slave can request a master/slave switch, to become the master itself
- slaves do not talk to each other

The BT protocol stack [239] To achieve interoperability, matching applications must run over identical protocol stacks. There must be a common data link and physical layer.

Protocol layers [255]:

- Bluetooth Core Protocols: Baseband, LMP, L2CAP, SDP
- Cable replacement protocol: RFCOMM
- Telephony control protocols:
- Adopted procotols:

Host Controller Interface (HCI) a command interface to the lower layers: baseband controller, link manager, HW status, control registers.

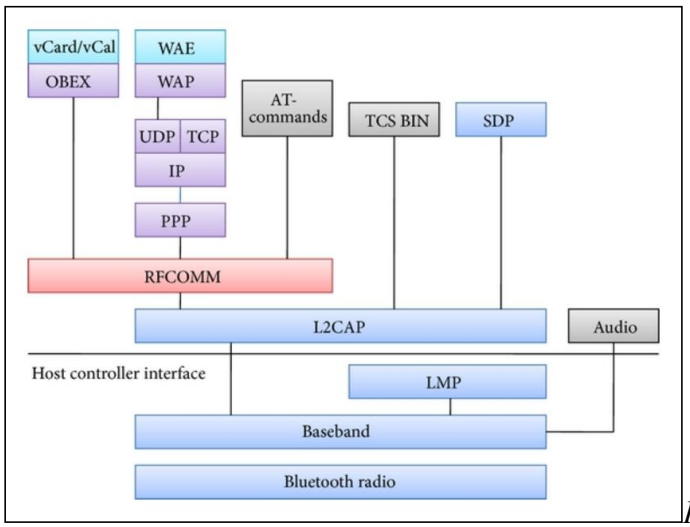


Figure 1: Bluetooth Protocol Stack model (Link)

Baseband and link control	Implements the physical link between BT devices, forming the piconet. It provides two physical links, ACL for data and SCO for audio or a combination of audio and data. Packets can have different levels of error correction and can be encrypted.
Link Management Protocol (LMP)	Responsible for link setting between BT devices.

L2CAP	**Logical Link Control and Adaptation Protocol**: it adapts upper layer protocols over the baseband. It works in parallel with LMP, when payload data is not sent at LMP. L2CAP provised connection and connectionless data serives to the upper layers. L2CAP allow to send/receive packets up to 64kB in length. The Baseband protocol provides SCO and ACL, but the L2CAP is defined only for ACL in Bluetooth 1.0 [315].
SDP	**Service Discovery Protocol**. It allows to query device information, services and service characteristics. Then, a connection between two or more BT devices can be establlished.
RFCOMM (*it emulates RS-232 as transport*)	The Cable Replacement Protocol. This protocol emulates RS-232 control and data signals over over Bluetooth baseband, providind transport capabilities for upper services.
TCS BIN	**Telephony Control Protocol – Binary**. A bit oriented protocol defining the call control signaling to establish speech and data calls between BT devices.
PPP	Point-to-Point Protocol. It takes IP packets to/from the PPP layer and places them onto the LAN.
OBEX	(IrOBEX) is a session protocol to exchange objects in a simple and spontaneous manner. Similar to HTTP but lighter, it uses a client-server model independent of the transport mechanism. OBEX provides a model for representing objects and operations. OBEX also defines folder-listing objects, to browse the content of folders in remote devices.
Content formats: vCard, vCalendar, vMessage and vNote.	Open specifications defining the format, then transported by lower layers.
WAP	

6.2 The Bluetooth Profiles

The BT SIG identified some usage models [404]. Each usage model is accompanied by a **Profile**. A BT Profile defines the protocol and protocol features supporting that usage model (*i.e. use case?*) There are also four general profiles:

- Generic Access Profile (GAP)
- Serial Port profile
- Service Discovery Application Profile (SDAP)
- Generic Object Exchange Profile (GOEP)

6.2.1 File Transfer

[413] Transfer data from one device to another: single file, folders, streaming media formats are supported. Remote folder browsing is supported. The required protocol stack includes OBEX and SDP.

6.2.2 Synchronization [422]

6.2.3 Three-in-One Phone

6.3 Bluetooth Specifications

Bluetooth 1.0 and 1.0B	[489] Early 2000. Not much adoption, because it was riddled with problems.
BT 1.1	June 2002.
BT 1.2	2005. Improved the transmission speed to up to 721kbit/s
BT 2.0 + EDR	2004. Enhanced Data Rate (EDR). Now theoretical speed of3Mbit/s, in practice 2.1Mbit/s.
BT 2.1 + EDR	2007
BT 3.0 + HS	2009. Featured the optional High Speed (HS) support. Improved the data speed, but not directly through the BT link. Alternate MAC/PHY (AMP): the transmission happended through an aternete 802.11 MAC layer.
BT 4.0 + LE	2010. The standard featured the optional Low-Energy (LE) support. BT LE is an entirely new protocol stack previously called Wibree. BT LE simplifies the discovery and pairing of simple links.
BT 4.1	2013 no new headline feature
BT 4.2	2014. Support for the Internet Of Things (IOT) usage.
BT 5	[559] June 2016. Speed increased to 2 Mbit/s

7 Inside Bluetooth Low Energy ([GU16])

IEEE 802.15 is the standard family that includes the Bluetooth [21].
Lower layers are referred to as the controller. BD_ADDR= Bluetooth Device Address [28]. Connection set-up: in case the Device A is scanning, and it want to connect with the device B, the connection process is also called paging. A is the Master, B is the Slave, they are connected. Network topologies: piconet (only master and slave). Scatternet: combination of two or more piconet. By default the device initiating the connection is the Master [41] during the connection, the Slave can become the master by executing a role switch.

7.1 Bluetooth Lower Levels

Security [60] procedures include: pairing, authentication, encryption, secure simple pairing.

- Pairing: the devices exchange a passkey. The passkey is also called PIN. Each device stores the passkey on its side [61]
- Authentication: verifying who is at the other side of the link. The verifier sends a challenge, the other side replyes with a response. The challenge-response is based on the previously exchanged passkey.
- Encryption: encryption is optional. Authentication must have been established to enable encryption [62].
- Secure simple pairing (**SSP**) simplifies the pairing and improves security [73]. SSP introduced 16 characters alphanumeric pin.

SSP is against passive eavesdropping and man in the middle attacks (**MITM**).

- Passive eavesdropping depends on the PIN key and the strong encryption mechanism [73].
- MITM to prevent MITM attacks, SSP provides:
  - Numeric Comparison
  - Passkey entry

7.1.1 Association Models

Association models depend on the I/O capabilities of the devices.

- Numeric Comparison: both devices can show a code and the user can say yes or no. It protects against MITM.
- Just Works: works without user intervention. It does not protect against MITM
- Out Of Band: it uses an external means for discovering the devices and exchanging pairing information.
- Passkey Entry: at least a device must have input capabilities, the other must have display capabilities. The passkey is shown on the device that has display capabilities, and the yes/no is entered on the device that has input capabilities.

8 Security Manager (SM)

Security breaches are due to passive eavesdropping and MITM attacks [232]. The Security Manager provides four types of security:

- LE secure connections pairing
- authenticated MITM protection: if a device has input capabilities and the other has isplay capabilities, one can show and the other can enter the key.
- unauthenticated no MITM protection: it is the Just Works pairing method
- no security: no support for authentication and encryption.

8.1 Pairing

[239] Three phases:

1. Feature exchange [239]
   - I/O capability: depending on the I/O capabilities, the proper pairing procedure is selected for the next phase. Input: None, Yes/No, keyboard; Output: None, Numeric Ouput
   - OOB authentication data
   - key size (the encryption key size is selected, not the passkey size)
   - Repeated attempts
2. Authentication and encryption: depending on the step 1, the method is chosen, among: Just Works, Numeric Comparison, Passkey Entry, OOB. Note [242]: if both devices have not set the MITM, then Just Works is selected.
3. Transport Specific Key distribution (optional)

The features must be set before the pairing begins.

Passkey Entry provides protection agains MITM, but not against eavesdropping [243].

[81]

9 Biography

ID	Title	Version
[GC19]	“Bluetooth Tutorial”, G.Colbach, 2019	
[GU16]	“Inside Bluetooth Low Energy”, Artech House, 2016.	2nd edition.
[KT14]	“Getting Started with Bluetooth Low Energy”, K.Townsend et al., 2014, O'Reilly	1st edition
[SJ16]	“Doing Bluetooth Low Energy on Linux”, Szymin Janc, link	2016
[UGEN]	UUID generation page: https://www.itu.int/en/ITU-T/asn1/Pages/UUID/uuids.aspx	

https://github.com/fbuihuu/samples-dbus

https://developer.gnome.org/gio/stable/gdbus-convenience.html